

20 Years and 20 Decimal Digits: A Journey With Optical Frequency Combs

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Abstract—Precision measurements represent a cornerstone in fundamental science. The capability of observing and quantifying subtle phenomena and events allows new discoveries and it confirms or confutes the theories describing our understanding of nature. The optical frequency comb, providing hundreds of thousands of phase-locked and evenly spaced laser lines, is one of the most fascinating enabling optical technologies and it is the result of a continuous pursuit for precision. Within the two decades from its inception, it has become a key instrument in many laboratories and has revolutionized numerous fields, spanning from time, frequency and length metrology to attosecond physics, gas-sensing and molecular fingerprinting. In this letter we summarize some steps of an exciting journey started 20 years ago, with a certain focus on the authors' contribution, finally leading to the demonstration of frequency measurements at the 20th decimal digit, and we show some prospective developments.

Index Terms—Frequency comb, time and frequency metrology, microwave synthesis, spectroscopy, frequency reference.

I. INTRODUCTION

“NEVER measure anything but frequency!” The legacy of Arthur Schawlow became the source of inspiration that opened the path to the conception of the optical frequency comb (OFC) [1]. Nowadays, frequency is the physical quantity that can be determined and synthesized with the highest degree of stability and accuracy. The comb was originally envisioned to ease the difficulty to determine optical frequencies against readily available radiofrequency references, especially the Cs ground-state hyperfine splitting determining the SI second [2]–[8]. In reverse, providing the possibility to count optical cycles, the OFC opens the pathway for realizing clocks based on optical frequency standards [3], [9], [10]. At the Max-Planck-Institute of Quantum Optics (MPQ) in Munich (and previously at Stanford) it has been the 1S-2S transition in atomic hydrogen that inspired many new developments. John Hall's group at JILA, the joint institute of the University of

Colorado Boulder and the National Institute of Standards & Technology joined the believers that combs were the technology that would have changed the world of optical frequency measurements [11], [12]. In the late 90s, researchers realized that the measurements were not limited by the spectroscopic apparatus, but by the ability to measure the excitation laser's frequency.

After extensive tests starting in 1996, researchers showed that the comb-like mode spectrum of a fs laser is an ideal evenly spaced frequency ruler [13], and a new path to frequency measurements was established. Concurrently, schemes for spectrally locating this frequency comb were sought, leading to the most prominent outcome of the f - $2f$ self-referencing method [3], [14]. In 1999, twenty years ago, the first optical frequency measurement of the D1 line of cesium using a frequency comb was published [2], and the first absolute frequency comb measurement of the 1S-2S H transition was conducted [4].

Only 2 years later, Menlo Systems was founded, determined to commercialize this technology and make it available to an increasing group of users. Exploiting commercial OFCs, many breakthroughs and advances in precision measurements and fundamental physics have been made possible, e.g. [15]–[17]. In this letter we take you through an exciting journey that started 20 years ago. During such period, this technology has been progressively advanced, for ruggedness, compactness, accuracy, and ease of operation. After a historical introduction describing the main source of inspiration that gave birth to the OFC, we outline some recent achievements: spectral purity transfer in the optical domain at the 20th digit of relative uncertainty for operating and comparing optical clocks, high-fidelity photonic microwave synthesis, difference-frequency generation (DFG) for mid-IR sources, space- and astro-comb development.

II. PROGRESS IN PRECISION FOR FUNDAMENTAL PHYSICS, CLOCKS, AND OSCILLATORS

The most popular approach to obtain an OFC is by using a mode-locked femtosecond laser. In the Fourier domain, the spectrum is composed by modes separated by $f_{rep} = 1/T$, where f_{rep} is the repetition rate of the coherently emitted laser pulses, given by the intra-cavity round-trip time $T = 2L/v_{gr}$ with cavity length L and group velocity v_{gr} . Since the phase and group velocity of the pulse may differ due to dispersion, a pulse-to-pulse phase slip $\Delta\phi$ is accumulated between the

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carrier frequency and the pulse envelope, corresponding to an offset from zero of the frequency spectrum by the so-called carrier-envelope-offset frequency $f_{ceo} = f_{rep} \times \Delta\phi/(2\pi)$. If the two quantities f_{rep} and f_{ceo} are known [18] (and, ideally, referenced through phase-locks), the frequency ruler is established and each laser mode is well-identified as $f_n = f_{ceo} + n \times f_{rep}$, with n being large integers. Through this relation, the OFC permits to count the fast cycles of optical transition frequencies in atomic systems, serving on one hand as very stable frequency references and on the other hand as window to the fundamental laws of nature.

Hydrogen is the simplest natural atomic system, and it represents a key tool for testing fundamental theories, in particular quantum electrodynamics (QED), since the dawn of modern physics. Already long before the '90s, scientists realized that determining the frequency of light instead of its wavelength would increase the accuracy in hydrogen spectroscopy. Complex frequency chain setups were maintained to allow the determination of optical frequencies by up-converting the unperturbed ground-state hyperfine transition frequency of the cesium 133 atom [19], [20]. These frequency converter chains did not offer the flexibility for addressing a wide optical frequency range, implying the need to greatly modify the setup whenever a new optical frequency measurement was planned. Optical frequency interval dividers had been developed to bridge larger gaps in these frequency chains [21]. The advent of the femtosecond laser frequency comb was a tremendous game changer. In fact, the ~ 2.466 PHz 1S-2S hydrogen transition frequency measurement accuracy stepped from 840 Hz before the advent of the comb [21] to 46 Hz right after [6]. Within a few years, progressive improvements were made, from 1.4 parts in 10^{14} in 2004 [22], to 4.2 parts in 10^{15} relative accuracy in 2011 [23]. Spectroscopy of other transitions allowed, *e.g.*, the indirect determination of the proton radius [24], which is at the center of an on-going challenge often addressed as the "proton radius puzzle" [25].

The 1S-2S dipole-forbidden transition frequency would be a computable frequency standard platform to construct an optical clock. Unfortunately, it remains extremely difficult to trap and laser-cool hydrogen. Consequently, other atoms and ions with narrow transitions are better suited for clock operation [9], [10], [26], even though the transition frequencies cannot be calculated from first principles. Over the course of the last two decades, optical frequency standards have progressed tremendously, now reaching the low 10^{-18} level of fractional stability and relative uncertainty, surpassing the ones of cesium fountain clocks by two orders of magnitude [26]–[29]. This fast development is linked to the advent of new concepts such as reduction of the quantum projection noise (QPN) limit by interrogating a large number of neutral atoms trapped in optical lattices [30], [31]. Moreover, tremendous technical progress on ultra-stable lasers (USL), *i.e.* continuous-wave lasers locked to high-finesse optical cavities, have led to coherence times beyond 50 s, corresponding to mHz-level linewidths and enabling high-precision spectroscopy on extremely narrow optical transitions [32]. Based on novel ideas for mechanical suspension [33] from the UK's National Measurement Institute NPL, Menlo Systems has developed small

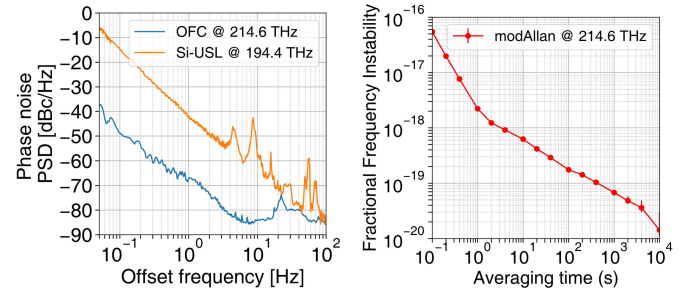


Fig. 1. Fidelity transfer using a rack-mountable ULN OFC. On the left we show the phase noise power spectral density of a USL reported in [31] (orange), and the residual phase noise of the ULN OFC (blue), obtained heterodyning two semi-identical systems when commonly referenced in the optical domain at 194 THz. On the right, the comb instability in terms of modified Allan deviation is shown.

transportable reference cavities for turnkey USLs, providing a complementary tool to the OFC. Recently, thanks to a close collaboration with Crystalline Mirror Solutions, who provides GaAs/AlGaAs semiconductor coatings [34], these USLs reach fractional frequency instability in the mid- 10^{-16} level at 1s. With the introduction of the robust “figure 9” mode-locking technology [35] and important improvements of OFC’s control methods [36], [37], fiber-based frequency combs have reached ultra-low noise (ULN) performance and now rival the best Ti:Sapphire-based OFCs. In fact, cycle-slip-free operation for weeks have pushed phase-coherent measurements to a new limit, with the ability to transfer the stability of the best reference cavities to far-spread optical frequencies. Fig.1 shows the optical phase noise power spectral density (PSD) of ULN comb lines at ~ 214 THz. The data have been derived heterodyning two self-referenced frequency combs, independently stabilized at $\nu_{ref} \sim 194$ THz, similarly to the setup described in [38]. Alongside, the resulting fractional frequency stability is displayed in terms of modified Allan deviation. Such level of performance reveals that the comb itself has a negligible contribution to the instabilities now reached by state-of-the-art optical clocks and 100-fold lower phase noise than the narrowest USL [32]. Due to low phase noise of the CEO frequency and the optical referencing, the stability of USLs can be transferred to all comb lines between at least 0 and $2\nu_{ref}$. With such spectral purity transfer from 194.4 THz up to 429.2 THz, QPN-limited stability on Sr optical lattice clocks was demonstrated at JILA [39]. Moreover, with a technique named dichroic heterodyne detection, optical frequency ratio measurements at the 20th digit stability and relative accuracy (outlined in detail in [38]) are now possible within minutes. For a future redefinition of the SI-second, clock comparisons in laboratories at different locations are of utmost importance. Fiber links and OFCs allow such inter- and intra-comparisons [40]. Outstanding progress is also made for free-space links [41], now reaching residual instabilities comparable to the one reached on fiber links [42].

The synthesis of narrow optical frequencies could serve well in the emerging research field of ion- or neutral atom-based quantum computing. In the quest to stabilize multiple light sources for laser cooling, quantum state preparation, quantum gates execution, and state detection, such universal OFC reference can be used as anchor for all involved lasers.

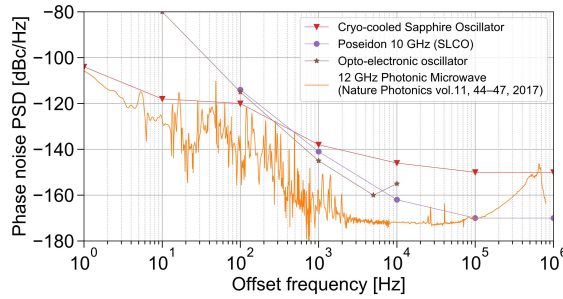


Fig. 2. Performance comparison of 12 GHz photonic microwave synthesis from [45] with other oscillator technologies for phase noise PSD.

Since the stability of modern optical references surpasses the stability of their radio-frequency counterparts, the coherent down-conversion from the optical to the microwave and RF domain is an intriguing option to create low-noise radiofrequencies and microwaves. As an example, such enhanced microwave probe frequency improves the performance of atomic fountain clocks [43], [44]. Many other applications could profit from ultra-stable microwaves [45], such as synchronization in large-scale accelerator facilities, telecommunications, opto-electronic sampling and radar surveillance. In 2016, in a joint effort with SYRTE (Paris), the synthesis of a 12 GHz microwave carrier with a timing noise floor at the zeptosecond level was derived from an optical reference cavity using a ULN frequency comb [46]. With such results, photonic microwave synthesis now surpasses in terms of phase noise the industrial standards [47]. Fig. 2 compares the results of [46] with typical phase noise performance of commercial opto-electronic oscillators, state-of-the-art cryo-cooled and room temperature sapphire-loaded cavity oscillators.

III. COMBS FOR APPLICATIONS OUTSIDE THE LAB

Many OFC applications require more robust and turnkey solutions, specifically to allow investigations outside the laboratory. A particularly challenging environment is space, where frequency combs could be used on satellites to operate optical clocks, ultimately improving the sensing of gravitation or the timekeeping for global navigation and positioning systems. Technology for such applications is strongly SWaP-constrained (i.e. limited in size, weight and power) and must additionally be very robust to survive the launch on a rocket. Together with different project partners, Menlo Systems has taken part in three projects that demonstrate the operation of automated frequency combs [48] and optical references [49] on sounding rocket flights, leading to a complete vacuum-compatible dual-comb with a volume of only 7 liters in 2018.

High precision spectroscopy in the molecular fingerprint region, e.g. for gas sensing applications, demands for OFCs with wavelength range in the mid-IR and in-field applications ask for reduced SWaP apparatuses, too. Menlo Systems has developed several schemes for DFG-based OFC systems allowing to access the mid-infrared region, as depicted in Fig. 3. Within the 3-5 μm region, turnkey Er/Yb:fiber systems with wide tuning range and high output power levels above 100 mW are available. When sacrificing some output power, a further extension into the range of 5-15 μm is possible. Combining 2 DFG systems, spectroscopy in the

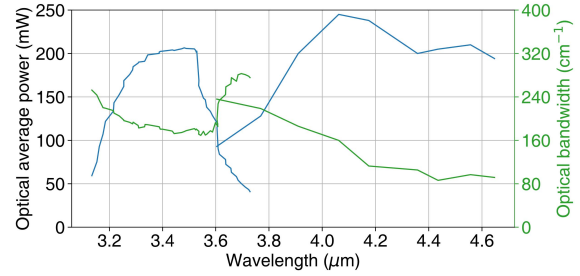


Fig. 3. Spectral coverage and bandwidth of the 3-5 μm region realized with Menlo Systems DFG Mid-IR rack-mounted OFCs.

mid-IR region can be addressed with the dual-comb technique [50], [51], allowing fast and accurate retrieval of molecular fingerprints.

A peculiar application of frequency combs outside the laboratory is the calibration of spectrographs at observatories, which has been proposed by MPQ and Menlo Systems, in close collaboration with the European Southern Observatory [52]. When hunting for maximum precision, these spectrographs need continuous calibration, which is traditionally done with spectral lines of thorium/argon lamps, unfortunately limiting the accuracy. However, accuracy over years is needed for several interesting investigations, such as the detection of Earth-like extrasolar planets via radial-velocity measurements. Calibrating the spectrograph by use of a frequency comb [53], the velocity of distant stars can be determined down to 2.5 cm/s through the Doppler shift of the star's spectrum [54]. Further applications are the direct measurement of the acceleration of the cosmic expansion, and a refined cosmological search for variability of fundamental constants. The challenge of such astro-combs is to provide a flat spectrum with appropriate mode spacing of up to 25 GHz, adapted to the spectrograph's spectral resolution and coverage.

IV. CONCLUSIONS AND OUTLOOK

We have highlighted some of the major achievements in OFC's technologies during these exciting last 20 years. We are looking towards the future being engaged in further challenging projects, and a steady quest to render frequency combs smaller and yet more robust. In 2007, microresonator-based OFCs were pioneered at MPQ, in collaboration with Tobias Kippenberg, now at EPFL (Lausanne) [55], [56]. First successful experiments have led to a completely new field of research, nowadays a trending research field [57], with the potential of complete photonic integration and turnkey operation.

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